

Start

1. Initialize Model Parameters

$$\mathcal{W} = \{w_1, w_2, \dots, w_n\}$$

2. Forward Propagation

$$\hat{y} = f(X; \mathcal{W})$$

3. Compute Loss

$$\mathcal{L}(\hat{y}, y)$$

4. Backpropagation

$$\nabla_{\mathcal{W}} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mathcal{W}}$$

5. Parameter Update

$$\mathcal{W}_{t+1} = \mathcal{W}_t - \eta \nabla_{\mathcal{W}} \mathcal{L}$$

Repeat until Convergence

